Kamyren Brown

Dr. Hubert Kyeremateng-Boateng

COSC 369.100

8 September 2026

Software Write Up: Summer Camp Lunch Package Registration

The Summer Camp Lunch Package Registration system will be a web application that allows parents and guardians to order lunches for campers attending summer camp at Tucker Road Ice Rink. The camp operates four days per week over a three-week period, giving each camper a total of 12 camp days. Parents will be able to order one lunch per camper for each selected day. The purpose of the system is to organize lunch registrations, calculate costs, and provide accurate meal totals to the café. This will help prevent missing orders, incorrect lunch counts, and unnecessary food waste.

Parents will have three package options. They can purchase lunches for the entire camp period, which covers 12 days and 12 lunches; half of the camp period, which covers six days and six lunches; or one week of camp, which covers four days and four lunches. For every covered day, the parent will select one of three meals. The chicken tenders, fries, and drink package will cost $11 per day. The cheese pizza, fries, and drink package will also cost $11 per day. The pepperoni pizza, fries, and drink package will cost $12 per day. The system will automatically calculate the total price based on the number and types of lunches selected.

The primary users will be parents or guardians, summer camp staff, café employees, and administrators. Parents will enter the camper's name, camp session, emergency contact information, allergies or dietary restrictions, selected package limit, camp dates, and meal choices. Through their account, parents will be able to track the number of lunch orders placed, the number of completed orders, and the number of days remaining in camp. Summer camp staff will be able to confirm each camper's registration and lunch status. Café employees will view the number of each meal that must be prepared for a specific day. The administrator will manage meal options, prices, camp dates, registrations, and order statuses.

The user interface will include a registration form, package selection page, daily meal selection page, order confirmation page, and parent dashboard. The parent dashboard will provide a clear summary of the camper's upcoming and completed lunches. Staff members will have a separate dashboard where they can search for campers and review or update orders. The café dashboard will show the number of chicken tender, cheese pizza, and pepperoni pizza meals required for the day. The interface will be simple and easy to understand because users may have different levels of experience with technology.

HTML and CSS will be used to create and style the application pages. JavaScript will add interaction, validate required information, update prices, and display order totals. Python and

Flask will handle the application's main logic, connect the user interface to the database, and process registrations and order updates. Flask is appropriate because it is a lightweight Python framework that can support forms, user accounts, multiple pages, and database operations without making the project unnecessarily complicated.

SQLite will store camper information, parent or guardian information, allergies, camp dates, package limits, daily meal selections, prices, and order statuses. It is suitable for this project because it can connect directly to Flask and does not require a separate database server. GitHub will be used to store the project document and code, track changes throughout development, and maintain earlier versions of the project. Together, these technologies will provide the user interface, database connection, order tracking, and other functions required for a complete software system.